

Sylvia Amihere

Postdoctoral Researcher

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Baltimore, MD

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Professional Experience

- Aug. **Postdoctoral Researcher**, *University of Maryland, Baltimore County*, Baltimore, MD.
2025–Present Mentor: Daniel R. Reynolds
- 2024–2025 **Postdoctoral Researcher**, *Southern Methodist University*, Dallas, TX.
Mentor: Daniel R. Reynolds

Education

- 2024 **PhD in Mathematics**, *The University of Alabama*, Tuscaloosa, AL.
Advisor: Shan Zhao
- 2024 **MA in Mathematics**, *The University of Alabama*, Tuscaloosa, AL.
- 2016 **BA in Mathematics with Economics**, *University of Ghana*, Accra, Ghana.

Research Interests

- Core Areas Temporally adaptive multirate infinitesimal (MRI) methods, SUNDIALS/ARKODE development, Poisson–Boltzmann solvers, Numerical PDEs, and Scientific Computing.

Research Experience

- 2024–Present **DOE Funded Projects**, *University of Maryland, Baltimore County*.
- Developing advanced techniques for temporal adaptivity in both single-rate and multi-rate systems of differential equations.
 - Developing strong-stability-preserving Runge–Kutta methods for gyrokinetic simulations.
- 2021–2024 **NSF Supported Projects**, *The University of Alabama*, DMS-2110914, DMS-2306991.
- Developed a new boundary treatment for the nonlinear Poisson–Boltzmann Equation (PBE).
 - Investigated numerical and regularization methods for solving the PBE.
 - Modified the rMIB Poisson–Boltzmann solver to accurately compute the electrostatic free energy of the nonlinear PBE.
- 2015–2016 **Undergraduate Research Thesis**, *University of Ghana*, EIGENFACES.
- Carried out face recognition using eigenfaces.
 - Coded eigenvectors of images using Python.
 - Investigated how lighting conditions affect face recognition.
- Fall 2014 **Undergraduate Research Project**, *University of Ghana*, KINETICS OF LEAD TRANSFER IN THE BODY.
- Investigated the transportation of lead in the human body using ordinary differential equations (ODEs) and Python programming.

Software Contributions

- SUNDIALS SUite of Nonlinear Differential and ALgebraic Solvers
- GX GPU-native nonlinear gyrokinetics

- rMIB-PB Regularized Matched Interface and Boundary Poisson–Boltzmann Solver (with Robin Boundary condition)
- MATLAB Nonlinear electrostatic free energy of a Kirkwood Sphere

Technical Skills

Languages Fortran90, Python, MATLAB, Experience with C++, C

Publications

- 2026 Maggul, M., Francisquez, M., Reynolds, D.R., **Amihere, S.** Super Time Stepping Methods for Diffusion using Discontinuous-Galerkin Spatial Discretizations. *arXiv:2026*.
- 2026 D. R. Reynolds, **S. Amihere**, D. Mitchell and V.T. Luan, Efficient and Flexible Multirate Temporal Adaptivity, *Journal of Computational and Applied Mathematics*.
- 2025 **S. Amihere**, Y. Ren, W. Geng, and S. Zhao, A new boundary condition for the nonlinear Poisson–Boltzmann equation in electrostatic analysis of proteins, *Journal of Computational Physics*, 528, 113844.
- 2024 Ren, Y., **Amihere, S.**, Geng, W., and Zhao, S. Comparison of three matched interface and boundary (MIB) schemes for solving the nonlinear Poisson–Boltzmann equation. *Communications in Information and Systems*, 24, 231-251.
- 2022 **S. Amihere**, W. Geng, and S. Zhao, Benchmarking electrostatic free energy of the nonlinear Poisson–Boltzmann model for the Kirkwood sphere, *Communications In Information and Systems*, 22(3), 305-315.

Awards & Honors

- Apr. 2024 **Outstanding Teaching Award**, *College of Arts and Sciences*, University of Alabama.
- Apr. 2024 **Outstanding Teaching Award**, *Department of Mathematics*, University of Alabama.
- Mar. 2024 **American Mathematical Society (AMS) Travel Award**, *Spring Southeastern Sectional Meeting*.
- Jan. 2024 **Joint Mathematics Meetings (JMM) Travel Award**, *American Mathematical Society*.
- Jan. 2024 **Graduate Student Travel Award**, *Graduate School*, University of Alabama.
- 2022-2024 **Summer Research Fellowship**, *Department of Mathematics*, University of Alabama.
- Nov. 2023 **Graduate Student Travel Award**, *Department of Mathematics*, University of Alabama.
- May 2022 **ICCMAE 2022 Travel Award**, *Mississippi State University*.
- May 2022 **Graduate Student Travel Award**, *Graduate School*, University of Alabama.
- Nov. 2016 **Best Graduating Female Student in Mathematics**, *University of Ghana*.

Talks & Presentations

- June 2026 **ICM 2026 Satellite Conference**, *Morgan State University*, Maryland.
- May 2026 **SMARTS-FRONTIERS All-Hands Meeting**, *University of California at San Diego*.
- May 2026 **SAYAS Numerics Day**, *University of Maryland, College Station*.
- May 2026 **Science Slam**, *Baltimore Underground Science Space*, BUGSS.
- Mar. 2026 **STEM Club - PI Day**, *Baltimore City Community College*, BCCC.
- Mar. 2026 **Differential Equations Seminar**, *Department of Mathematics & Statistics*, UMBC.
- Nov. 2025 **Mid-Atlantic Numerical Analysis Day**, *Department of Mathematics*, Temple University.

- Oct. 2025 **Finite Element Circus**, *Center for Mathematics and Artificial Intelligence*, George Mason University.
- Apr. 2025 **Applied Math Seminar**, *Department of Mathematics*, University of Ghana.
- Apr. 2025 **2025 Rising Stars in Computational and Data Sciences**, *ODEN Institute*, UT Austin, Texas.
- Mar. 2025 **2025 SIAM Conference on Computational Science and Engineering (CSE25)**, *Fort Worth*, Texas.
- Oct. 2024 **Clements Scientific Computing Seminar Series**, *Southern Methodist University*.
- Mar. 2024 **AMS 2024 Spring Southeastern Sectional Meeting**, *Florida State University*.
- Jan. 2024 **Joint Mathematics Meetings (JMM) 2024**, *Moscone Center*, San Francisco, CA.
- Nov. 2023 **The 6th SIAM Texas-Louisiana Sectional Meeting 2023**, *University of Louisiana at Lafayette*.
- Oct. 2023 **Mathematics Graduate Student Seminar**, *The University of Alabama*.
- Sept. 2023 **Graduate Students' Mathematical Fair 2023**, *Ghana Numerical Analysis*.
- Nov. 2022 **34th Joint Applied Math Meeting**, *University of Alabama in Huntsville*.
- May 2022 **The Second International Conference on Computational Methods and Applications in Engineering (ICCMAE)**, *Mississippi State University*.
- April 2022 **Online Seminar on Mathematical Molecular Bioscience and Biophysics (MMBB)**, *Michigan State University*.
- Feb. 2022 **Mathematics Graduate Student Seminar**, *The University of Alabama*.

Teaching Experience

- Spring 2026 **Instructor of Record**, *University of Maryland, Baltimore County*.
- Calculus and Analytic Geometry I
- 2018–2024 **Graduate Teaching Assistant**, *The University of Alabama*.
- Tutor, Mathematics Technology and Learning Center (MTLC)
 - Instructor for Linear Algebra (Math 237)
 - Instructor for Calculus I (Math 125)
 - Instructor for Differential Equations (Math 238)
 - Instructor for Finite Mathematics with applications (Math 110)
 - Instructor for Business Calculus (Math 121)
 - Instructor for Precalculus and Trigonometry (Math 115)
- 2016–2017 **Teaching Assistant**, *University of Ghana*.
- Organized tutorials to explain lecture notes to undergraduate students.
 - Assisted students with computational mathematics team projects.
 - Invigilated and graded interim departmental assessment tests.

Industry Experience

- 2017–2018 **Mathematics Project Assistant**, *Deffor Ventures*.
- Created interactive online mathematics lessons and curriculum exercises.
 - Reviewed online advanced mathematics questions and technical solutions.
- Fall 2016 **Mathematics Tutor**, *Legacy Tutor*.
- Prepared structured mathematics lesson notes for remote web delivery.
 - Provided target online tuition sessions for students struggling in basic mathematics.

- 2012–2016 **Administrative Assistant**, *Ange Dior*.
- Served as primary front-desk receptionist and handled initial client contact.
 - Assisted operational leadership in planning and coordinating large corporate events.

Mentorship & Professional Service

- 2025–Present **Journal Reviewer**, *Applied and Computational Mathematics*, Science Publishing Group.
- 2024–2025 **Academic Judging & Evaluation**, *National and Regional Competitions*.
- **M3 MathWorks Math Modeling Challenge**: Evaluated high school research manuscripts.
 - **Dallas Regional Science and Engineering Fair Academy**: Judged competitive middle and high school STEM projects.
 - **Ohio Academy of Science State Science Day**: Evaluated pre-college research methodologies and project accuracy.
 - **Undergraduate Research and Creative Activity Conference (UA)**: Evaluated visual poster presentations.
 - **Joint Mathematics Meetings (JMM)**: Provided formal feedback on undergraduate research posters.
- 2022–2023 **Graduate Mentor**, *The University of Alabama*, Department of Mathematics.
- Guided and co-mentored two Master's students through the completion and defense of their final theses.
 - Provided technical codebase guidance on converting legacy FORTRAN packages into modern MATLAB.

Leadership

- Mar. 2025 **Minisymposium Co-Organizer**, *SIAM CSE 2025*, Fort Worth, TX.
- Co-organized the special session “Mathematics in Structural Biology” for the SIAM Conference on Computational Science and Engineering.
- 2022–2024 **UA Student Chapter Secretary**, *Association for Women in Mathematics*, Tuscaloosa, AL.
- Coordinated chapter events, managed inter-organizational communications, and authored monthly newsletters.
- Sept. 2023 **Co-Organizer**, *Graduate Students' Mathematical Fair*, Ghana Numerical Analysis.
- Organized and moderated a global virtual conference for STEM graduate students of African origin.
- 2015–18 **Founding Member**, *iVolunteer Network Ghana*.
- Provided community mathematics tutoring and managed cross-organizational financial documentation.

Community Service

- Jan. 2026 **Science Olympiad at UMBC**, *Grader*, Baltimore, MD.
- Jan. 2026 **Woodlawn High School Lunch & Learn**, *Volunteer*, Baltimore, MD.
- Oct. 2025 **Baltimore Polytechnic Institute**, *Outreach Volunteer*, Baltimore, MD.
- 2022–2024 **MathCounts Competition**, *Grader*.
- 2022, 2024 **JMM Graduate School Fair**, *Volunteer Outreach*.
- 2022, 2023 **UA GTA Workshop Team**, *Teaching Fellow*, Tuscaloosa, AL.
- Apr. 2023 **Troy MathFest Graduate School Fair**, *Volunteer*, Troy, AL.
- 2021–2022 **UA African Students Association**, *Community Volunteer*, Tuscaloosa, AL.

Professional Associations

- American Mathematical Society (AMS)
- Mathematical Association of America (MAA)
- Association for Women in Mathematics (AWM)
- Society of Industrial and Applied Mathematics (SIAM)